

Name: _____

These questions assess your understanding of the material from Chapters 1, 2 of Silberberg. Please write your name on the sheets of paper that you use and show all your work.

1. An engineer measures the capacity of an electricity-generating plant. The capacity is 496 MW. What is the capacity in watts? Report your answer in scientific notation.

ANSWER: _____

2. Provide the name and empirical formula of an ionic compound that is formed by the iron (II) cation and the fluoride anion.

ANSWER: _____

3. Write the name of the inorganic acid HClO.

ANSWER: _____

4. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$14.12 \frac{\text{mol}}{\text{L}} \times 0.985 \text{ L}$$

ANSWER: _____

5. A civil engineer measures the length of an airport runway. The length is 7164 m. What is the length in kilometers? Report your answer in scientific notation.

ANSWER: _____

6. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 2.41×10^{-14} m. What is the wavelength in femtometers? Report your answer as a decimal.

ANSWER: _____

7. Determine whether caesium and bromine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form.

ANSWER: _____

8. An optician measures the diameter of her patient's pupil. The diameter is 0.0072 m. What is the diameter in millimeters? Report your answer as a decimal.

ANSWER: _____

9. Provide the chemical formula for the molecular compound silicon tetrafluoride.

ANSWER: _____

10. Provide the chemical symbol of the most commonly formed ion of beryllium and indicate whether it is a cation or an anion.

ANSWER: _____

11. A geologist measures the weight of an ore sample. The weight is 5.8649 kg. What is the weight in grams? Report your answer in scientific notation.

ANSWER: _____

12. Efe must note the concentration of a solution of $(\text{H}_3\text{O})_2\text{CO}_3$ in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Efe weighs an empty graduated cylinder and finds its mass to be 6. g. He then puts some solid $(\text{H}_3\text{O})_2\text{CO}_3$ into the graduated cylinder and weighs it. With the $(\text{H}_3\text{O})_2\text{CO}_3$ added, the cylinder weighs 48.4 g. He adds water to the graduated cylinder and dissolves the $(\text{H}_3\text{O})_2\text{CO}_3$ completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 57.95 mL. What concentration should Efe write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

13. A millworker measures the width of a plank. The width is 2.2 dm. What is the width in meters? Report your answer as a decimal.

ANSWER: _____

14. Robert needs to know the mass of a sample of $(\text{NH}_4)_2\text{O}$ to 3 significant figures. He puts the sample on a digital scale. This is what the scale shows: 56.73 g. If this measurement is precise enough for Robert, round it to 3 significant figures and report your answer in scientific notation.

ANSWER: _____

15. A shipwright measures the length of a bolt. The length is 85.9 cm. What is the length in meters? Report your answer in scientific notation.

ANSWER: _____

16. A nuclear chemist measures the mass of an artificial element. The mass is 53 pg. What is the mass in grams? Report your answer in scientific notation.

ANSWER: _____

17. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$177. \text{ mol} \times 33.765 \text{ L}$$

ANSWER: _____

18. Classify P_4Si_3 as either an ionic or a molecular compound.

ANSWER: _____

19. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is $5.4786 \times 10^{-7} \text{ m}$. What is the wavelength in nanometers? Report your answer in scientific notation.

ANSWER: _____

20. Write the chemical formula for hydrosulfuric acid.

ANSWER: _____

21. Provide the element symbol that represents arsenic.

ANSWER: _____

22. Provide the name and empirical formula of the ionic compound that is formed by the Cu^+ cation and the $\text{Cr}_2\text{O}_7^{2-}$ anion.

ANSWER: _____

23. A lab technician measures the weight of a reagent. The weight is 3930 g. What is the weight in kilograms? Report your answer in scientific notation.

ANSWER: _____

24. A geologist measures the height of a mountain. The height is 3.7251 km. What is the height in meters? Report your answer in scientific notation.

ANSWER: _____

25. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is $7.76 \times 10^{10} \text{ m}$. What is the distance in gigameters? Report your answer in scientific notation.

ANSWER: _____

26. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 445 GHz. What is the frequency in hertz? Report your answer in scientific notation.

ANSWER: _____

27. As part of his duties as an analytical chemist, Jeremy measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. He then records these masses in a table in his lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Jeremy would find in a new 10.0-g sample of X. Round your answer to two significant figures.

sample	mass of E_1	mass of E_2
1	5.2 g	2.6 g
2	9.3 g	5.6 g
3	5.2 g	5.2 g
4	6.3 g	2.4 g

ANSWER: _____

28. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.339 m. What is the thickness in centimeters? Report your answer as a decimal.

ANSWER: _____

29. As part of his duties as an analytical chemist, Brian analyzes a sample of an unknown pure substance X and finds that it contains 5.6 g of element E_1 and 8.2 g of element E_2 , and no other elements. He then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in his lab notebook, reproduced below. Which of the other three substances is also a pure substance?

sample	mass of E_1	mass of E_2
X	5.6 g	8.2 g
A	16.1 g	11.4 g
B	15.2 g	11.1 g
C	14.8 g	14.0 g

ANSWER: _____

30. Provide the number of protons and the number of neutrons there are in an atom of helium-4.

ANSWER: _____

31. A geophysicist measures the pressure in a borehole. The pressure is 5.084×10^7 Pa. What is the pressure in megapascals? Report your answer as a decimal.

ANSWER: _____

32. A botanist measures the thickness of a leaf. The thickness is 2.0 mm. What is the thickness in meters? Report your answer as a decimal.

ANSWER: _____

33. Answer the following questions: is Al an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

34. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 1.79×10^{-5} g. What is the mass in micrograms? Report your answer in scientific notation.

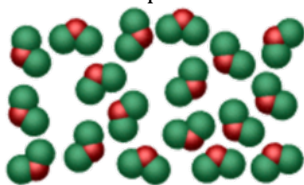
ANSWER: _____

35. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$19.622 \text{ mL} - 9.37 \text{ mL}$$

ANSWER: _____

36. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture?



ANSWER: _____

37. Provide the name for the molecular compound SF_6 . If the compound has a common name, you may provide that instead.

ANSWER: _____

38. A nurse measures the weight of his patient's medicine. The weight is 0.0082 g. What is the weight in milligrams? Report your answer in scientific notation.

ANSWER: _____

39. A cell biologist measures the width of a cell. The width is 66 μm . What is the width in meters? Report your answer in scientific notation.

ANSWER: _____

40. Write the chemical formula of rubidium bicarbonate tetrahydrate.

ANSWER: _____

41. Provide the name of the element that the element symbol Ga represents.

ANSWER: _____

42. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$11.5 \text{ mL} + 5.14 \text{ mL}$$

ANSWER: _____

43. A physical chemist measures the width of a molecule. The width is 31 nm. What is the width in meters? Report your answer in scientific notation.

ANSWER: _____

44. A pharmacist measures the mass of a powdered drug. The mass is 3133 mg. What is the mass in grams? Report your answer in scientific notation.

ANSWER: _____

45. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 55 fs. What is the time in seconds? Report your answer in scientific notation.

ANSWER: _____